

Types of Variables

Ages 10-14 · Independent, dependent and controlled variables — and how a control group is different.

Name _____ Date _____

Independent variable (I) — the one factor you change on purpose.

Dependent variable (D) — the factor you measure to see the effect.

Controlled variable (C) — a factor you keep the same in every trial, so the test is fair.

Control group (G) — a separate group that gets no treatment, used as a baseline to compare against.

Part A — Sort the factors

Experiment: Does plant food make bean plants grow taller? Four groups of bean plants get 0 g, 2 g, 4 g or 6 g of plant food each week. Everything else is kept the same for all the plants. After three weeks, every plant's height is measured.

Write I, D, C or G in each box.

- ☐ 1. The amount of plant food each week
- ☐ 2. The height of the plants after three weeks
- ☐ 3. The amount of water each plant gets
- ☐ 4. The type of bean plant
- ☐ 5. The group given 0 g of plant food
- ☐ 6. The size of the pots
- ☐ 7. How much light the plants get
- ☐ 8. How long the test lasts (three weeks)

Part B — Find the variables in each hypothesis

1. If the ramp is higher, then the toy car will roll farther.

Independent: _____ **Dependent:** _____

2. If a plant gets more hours of light each day, then it will grow more leaves.

Independent: _____ **Dependent:** _____

3. The warmer the water, the faster salt will dissolve in it.

Independent: _____ **Dependent:** _____

4. If you exercise for longer, then your heart rate will be higher.

Independent: _____ **Dependent:** _____

Part C — Controlled variable or control group?

Circle the right answer for each statement.

1. Every pot gets exactly the same amount of water.

Controlled variable Control group

2. Some plants get no plant food, so the others can be compared with them.

Controlled variable Control group

3. All the plants are kept in the same room.

Controlled variable Control group